

Value Change Identification Module (VCIM)

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Canonical Definition

Value Change Identification Module (VCIM) is the first Core Module of the Value Change Standard (VCS) within VFM® — Value Flow Mechanisms Architecture.

VCIM governs the identification of a potential or confirmed material change affecting existing governed value by establishing the reference Value State, observing the subsequent value condition, detecting a relevant difference, and determining whether that difference constitutes a legitimate Value Change Candidate.

The Module establishes the first controlled transition from:

Governed Value State

to

Identified Value Change.

VCIM does not determine the final type, magnitude, or causal basis of the change.

Its foundational task is:

to determine whether something material has happened to the governed value that requires formal Value Change governance.

Architectural Position

VCIM is the first module within VCS.

It receives the governed value basis established through:

VSS — Value State Standard

and precedes:

VCTM — Value Change Type Module.

Its architectural position is:

Governed Value State → VCIM — Value Change Identification → VCTM — Value Change Type → VCMM — Value Change Magnitude → VCCM — Value Change Causation Link → VCRM — Value Change Record

VCIM therefore establishes whether a governable change exists before later modules determine what kind of change occurred, how significant it was, what may have caused it, and how it must be recorded.

Internal Module Flow

Governed Value State → Reference State Selection → Observed Value Condition → Difference Detection → Context Preservation → Materiality Screening → Change Candidate Identification → Change Identification Determination → Identified Value Change → VCTM — Value Change Type Module

Operational Space

VCIM governs the point at which an existing Value State is examined for evidence of material change.

Its operational space includes:

governed value identification; reference-state selection; reference-point identification; observation of subsequent value condition; difference detection; temporal comparison; contextual comparison; beneficiary-aware comparison; capability change detection; utility change detection; operational change detection; economic change detection; physical change detection; digital change detection; strategic change detection; materiality screening; change candidate creation; change identification determination;

uncertainty affecting identification; false-change prevention.

Reference State Principle

A Value Change cannot be identified meaningfully without a reference.

VCIM therefore requires identification of the governed condition against which the observed value condition is being compared.

The reference may be:

Previous Value State

Value State Record

Governed Baseline

Defined Reference Date

Reference Event

Reference Condition

Reference Capability

or another explicitly governed comparison basis.

The foundational rule is:

No Governed Reference → No Governed Value Change Identification

This prevents arbitrary claims that value has increased, decreased, deteriorated, improved, or otherwise changed without identifying what the comparison is actually against.

Observed Value Condition

VCIM distinguishes between:

Reference Value State

and

Observed Value Condition.

The Observed Value Condition represents the materially relevant condition of the governed value at the comparison point.

It may be derived from:

operational observations; system measurements; financial information; market information; physical inspection; performance data; capability assessment; utilization information; mission data; infrastructure monitoring; digital-system telemetry; contractual developments; environmental information; validated external information; other contextually appropriate evidence.

VCIM does not require one universal observation method.

The observation method must be appropriate to the value being governed.

Difference Detection

VCIM identifies whether a meaningful difference exists between:

Reference Value State

and

Observed Value Condition.

The difference may concern:

amount; capability; utility; availability; productivity; accessibility; relevance; condition; performance; scarcity; strategic importance; operational usefulness; beneficiary relationship; economic significance; physical condition; digital capability; resource requirement.

Difference detection is not yet full Value Change characterization.

It establishes that a potentially governable delta exists.

Difference ≠ Value Change

VCIM must prevent every detected variation from automatically becoming a Value Change.

A difference may arise from:

measurement noise; expected operational fluctuation; temporary observation error; seasonal variation; normal market movement; incomplete information; inconsistent measurement basis; different observation methods; different beneficiaries; changed contexts; data latency.

Therefore:

Detected Difference $\neq$ Identified Material Value Change

VCIM requires the difference to pass a materiality screen before it enters formal downstream Value Change governance.

Materiality Screening

VCIM performs an initial determination of whether the detected difference is sufficiently material to become a Value Change Candidate.

Materiality may depend upon:

absolute difference; relative difference; percentage change; operational consequence; capability consequence; beneficiary consequence; strategic significance; persistence; reversibility; risk consequence; resource consequence; economic consequence; mission consequence; defined organizational threshold.

The materiality basis must remain linked to the governed Value Context.

A CHF 1,000 change may be immaterial in one context and decisive in another.

Likewise, a 1% capability loss may be insignificant for one system and mission-critical for another.

Change Candidate

Where a relevant difference passes the applicable materiality screen, VCIM creates a:

Value Change Candidate.

A Value Change Candidate represents:

a sufficiently supported indication that existing governed value may have materially changed and should proceed into formal Value Change characterization.

The candidate should identify at minimum:

Governed Value

Reference State

Observed Condition

Detected Difference

Materiality Basis

Observation Point

Relevant Context

Affected Beneficiary where applicable

Identification Confidence or Uncertainty

The candidate does not yet establish full change type, magnitude, or causation.

Identification States

VCIM may assign an explicit identification state.

1. No Material Change Identified

A difference is absent or remains below the applicable materiality basis.

2. Potential Value Change

A relevant difference is observed, but additional information is required before material change can be established.

3. Material Value Change Identified

Available information sufficiently supports the conclusion that a material change to governed value has occurred.

4. Change Identification Suspended

A determination cannot currently proceed because material information is missing, contradictory, unreliable, or under review.

5. Change Identification Invalidated

A previously suspected change is shown to result from measurement error, reference mismatch, contextual mismatch, duplicated information, or another non-value-change condition.

These states prevent uncertainty from being silently converted into certainty.

Context Preservation

Value Change identification must preserve the Value Context established upstream.

A change identified for:

Beneficiary A

does not automatically establish the same change for:

Beneficiary B.

Likewise:

Value Change in Context A

does not automatically mean:

Value Change in Context B.

VCIM therefore preserves, where materially relevant:

value-bearing object; beneficiary; purpose; reference framework; operational context; temporal context; geographic context; system context; relevant conditions.

This prevents context drift during Value Change analysis.

Temporal Comparison

Value Change is frequently identified across time.

VCIM may compare:

Value at T1

with

Value Condition at T2.

But temporal sequence alone does not establish change.

The comparison basis must remain sufficiently equivalent.

For example, comparing asset productivity during:

normal production

with productivity during:

scheduled shutdown

could create an artificial Value Change if the contextual difference is ignored.

VCIM therefore requires time and context to be evaluated together.

Comparable Basis Principle

A legitimate Value Change comparison requires sufficiently comparable reference and observation bases.

The following should not be silently compared as equivalent:

Different Value Definitions

Different Beneficiaries

Different Measurement Methods

Different Units

Different Reference Frameworks

Different Operational Conditions

Different Object Configurations

unless those differences are explicitly governed.

The principle is:

Difference in Measurement Basis $\neq$ Difference in Value

Positive and Negative Change Candidates

VCIM may identify candidates indicating:

Potential Positive Value Change

or

Potential Negative Value Change.

However, VCIM does not perform the definitive change-type classification.

That belongs to VCTM.

VCIM only preserves the observed directional signal where it is useful for downstream governance.

Cumulative Change Detection

Material Value Change may emerge through accumulation.

Individual observations may remain below the materiality threshold while their cumulative effect becomes significant.

VCIM therefore supports:

Difference 1

Difference 2

Difference 3

Difference n

→ Cumulative Material Change Candidate

This is particularly important for:

industrial deterioration; infrastructure aging; efficiency loss; technological obsolescence; resource depletion; capability degradation; data deterioration; maintenance burden; gradual productivity improvement.

A governance system that evaluates only isolated events may systematically fail to identify slow Value Change.

Multi-Dimensional Change Identification

A value-bearing object may experience simultaneous changes across several value dimensions.

For example:

Industrial Asset

may exhibit:

Physical Condition ↓

Energy Efficiency ↓

Maintenance Requirement ↑

Production Capability ↓

while:

Market Scarcity ↑

may increase another dimension of value.

VCIM should preserve these different Value Change Candidates rather than forcing them prematurely into one net value figure.

This allows later modules to characterize each change appropriately.

Uncertainty Preservation

VCIM must preserve material uncertainty affecting identification.

Uncertainty may arise from:

incomplete observation; measurement uncertainty; conflicting evidence; uncertain reference state; uncertain beneficiary impact; insufficient temporal information; changing context; delayed information; proxy measurement; uncertain value definition.

Where uncertainty prevents legitimate determination, the result should remain:

Potential Value Change

rather than being promoted to confirmed change.

False Change Prevention

VCIM should actively distinguish real Value Change from apparent change produced by methodological inconsistency.

False change may arise through:

measurement method change

unit conversion error

reference mismatch

data correction

different valuation methodology

beneficiary substitution

context substitution

object version mismatch

duplicate observation

system migration

currency conversion effects

where those effects do not represent the governed Value Change being examined.

This is essential in automated environments where large data streams may otherwise create large numbers of false Value Change events.

Automated Change Identification

VCIM can support automated monitoring.

A machine-readable implementation may operate as:

Governed Value State

→ Continuous or Periodic Observation

→ Difference Detection

→ Materiality Screen

→ Value Change Candidate

→ Human or Automated Determination

→ Downstream Change Governance

Potential implementations include:

banking monitoring systems; asset-management platforms; digital twins; industrial IoT; blockchain analytics; infrastructure monitoring; insurance systems; AI resource governance; mission-control systems; public-asset management; automated accounting interfaces.

Automation must preserve the governed reference and context.

A system that detects numerical movement without understanding what the number represents is not yet governing Value Change.

Relationship to ZVA™

VCIM may identify a material change indicating that governed value is moving:

toward ZVA™ — Zero Value Asset™

or

away from ZVA™.

However, VCIM does not assign the resulting Value State.

For example:

Value State: Active Value

→ VCIM identifies material degradation

→ downstream VCS governance characterizes the change

→ VSS may subsequently determine ZVA™ where the applicable state conditions are satisfied.

Likewise:

ZVA™

may experience a material positive change that creates a candidate for a new non-zero Value State.

This preserves:

Change Identification ≠ State Determination

VCIM Boundary with VSS

VSS establishes:

What is the current governed Value State?

VCIM establishes:

Has something materially changed relative to that governed state?

Therefore:

Value State ≠ Value Change Identification

VCIM consumes the state as a reference but does not redefine the state itself.

VCIM Boundary with VCTM

VCIM determines whether a material Value Change has been identified.

VCTM determines what kind of Value Change occurred.

Therefore:

Identification ≠ Classification

VCIM must not prematurely assign the final Change Type.

VCIM Boundary with VCMM

VCIM may require a preliminary materiality screen.

VCMM performs the formal determination of Value Change magnitude.

Therefore:

Materiality Screening ≠ Change Magnitude Determination

A change can be sufficiently material to enter governance before its precise magnitude is established.

VCIM Boundary with VCCM

VCIM identifies that value changed.

VCCM governs the relationship between the Value Change and potential causal events, conditions, activities, or mechanisms.

Therefore:

Change Identification $\neq$ Causation

VCIM must not require complete causal knowledge before a legitimate Value Change can be identified.

VCIM Boundary with External Systems

External systems may provide observations used to identify Value Change.

Examples include:

Market System $\rightarrow$ Price Information

Banking System $\rightarrow$ Payment Information

Accounting System $\rightarrow$ Asset Information

Industrial System $\rightarrow$ Performance Information

Blockchain $\rightarrow$ Transaction or State Information

IoT System $\rightarrow$ Physical Condition Information

Mission System $\rightarrow$ Capability Information

VCIM may consume these signals.

It does not automatically govern the systems that produced them.

Minimum Implementation Framework

A minimum VCIM implementation should establish:

1. Governed Value Identification

Identify the governed value being monitored.

2. Reference State

Identify the applicable governed Value State or other explicit reference.

3. Observation Basis

Define the information used to observe the subsequent value condition.

4. Comparability Check

Determine whether the reference and observed conditions are sufficiently comparable.

5. Difference Detection

Identify any materially relevant difference.

6. Context Preservation

Preserve beneficiary, purpose, reference framework, object, and relevant operational context.

7. Materiality Screening

Determine whether the difference is sufficiently significant to enter formal Value Change governance.

8. Uncertainty Assessment

Record material uncertainty affecting the identification.

9. Identification State

Assign the applicable Value Change Identification State.

10. Change Candidate Creation

Create the governed Value Change Candidate for downstream characterization where appropriate.

Governance Outputs

VCIM may produce:

Value Change Observation Record Value Change Reference Record Value Comparison Basis Value Difference Record
Value Change Materiality Screen Value Change Context Record Value Change Uncertainty Record Value Change
Identification State Value Change Candidate Material Value Change Identification No-Material-Change Determination
Change Identification Invalidation Record

These outputs provide the governed input for:

VCTM — Value Change Type Module.

Use Case 1 — Banking: Early Collateral Value

Change Detection

A bank finances a production facility.

The facility's last governed Value State indicates:

high productive capability

stable operating availability

acceptable energy efficiency

and

strong collateral usefulness.

Over time, operational information begins showing:

production throughput ↓

unplanned downtime ↑

energy consumption per unit ↑

maintenance interventions ↑.

No loan payment has yet been missed.

No formal accounting impairment has necessarily been recorded.

The market price may not yet have materially moved.

VCIM compares the new operational condition with the governed reference Value State.

Instead of waiting for a financial consequence, the bank can identify:

Potential Material Value Change

and create a Value Change Candidate for downstream analysis.

VCTM can then determine the type of change.

VCMM can determine its magnitude.

VCCM can examine relevant causal links.

The practical advantage is significant:

the institution can begin governing deterioration in the underlying value before that deterioration necessarily appears as default, accounting loss, or market repricing.

Use Case 2 — Space Mission: Detecting Value

Created by Resource Optimization

A spacecraft has a governed computational-resource Value State.

Mission software is then optimized.

The same mission-critical computation can now be performed using:

30% less computational capacity

while preserving the required operational result.

There may be:

no transaction

no market price

no ownership change

and

no accounting recognition.

Yet the mission now possesses additional usable computational capacity.

VCIM compares:

Previous Computational Resource State

with

Observed Post-Optimization Condition.

It identifies a material difference:

more mission capability can now be produced from the same computational resource base.

VCIM therefore creates a:

Positive Value Change Candidate.

The later VCS modules can characterize and quantify the change.

This demonstrates a fundamental VFM® capability:

Value Change can be identified directly from changes in operational usefulness and capability without requiring monetary expression.

Enterprise Application

VCIM provides organizations with a systematic method for answering one deceptively simple question:

Has the value actually changed?

This question can be applied across:

banking; insurance; investment; industrial production; infrastructure; energy; blockchain; digital assets; intellectual property; AI systems; computational resources; public assets; mission systems; natural resources; organizational capability.

The Module creates a common governance mechanism through which different systems can detect material Value Change without forcing all forms of value into the same measurement model.

Foundational Principles

Value Change requires a governed reference.

Difference is not automatically Value Change.

Materiality determines whether a difference enters formal Value Change governance.

Change identification does not require complete causal knowledge.

Change identification is not change classification.

Materiality screening is not magnitude determination.

A change in measurement method must not be mistaken for a change in value.

Context-specific change must not automatically become universal change.

Beneficiary-specific change must remain beneficiary-specific.

Cumulative small differences may become a material Value Change.

Value Change may be identified without monetary expression, transaction, accounting recognition, or market repricing.

Canonical Closing Statement

Value Change Identification Module establishes the entry control point of the Value Change Standard by determining whether existing governed value has materially changed relative to an explicit and sufficiently comparable reference state. Through governance of reference selection, observation, difference detection, context preservation, comparability, materiality screening, cumulative change, multi-dimensional change, uncertainty, false-change prevention, identification states, and Value Change Candidate creation, VCIM prevents ordinary variation, measurement inconsistency, contextual mismatch, or unsupported interpretation from being treated as genuine Value Change. By separating change identification from change type, magnitude, causation, resulting Value State, price movement, transaction, accounting recognition, and monetary expression, VCIM provides VFM® with the methodological gateway required to observe Value Change across physical, digital, economic, operational, strategic, financial, and future value systems. Before a system can determine what kind of change occurred or how large that change was, it must first establish that value actually changed.

Modul2

Governed Links

→ Value Change Standard (VCS)

→ VFM® Architecture Map

→ Next: Value Change Type Module (VCTM)

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